Certified Local Bounds for $O(4)$ Mass Asymptotics

Positive Integral Residuals, Spectral Moments, and Conditional Lattice Matching

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Abstract

We examine the precise normalization and dependency structure of OpenAI's October 5, 2026 manuscript on the two-dimensional $O(4)$ lattice mass, at the specified repository pin. Two local additions are proved. First, its odd dressed-energy integral equation admits explicit residual-to-coupling and residual-to-positive-tail-mass enclosures, including a truncated-tail certificate in exponentially scaled units. Second, two positive spectral traces give sharper norm bounds than the one-sided trace-doubling estimate used in the source; an optimal lower bound from spectral purity and a seam-error enclosure are also derived. Actual private Ram passages on finite local moments, residual stability, and positive moment identification are used with their original limitations. Neither polynomial moment coordinates nor successful finite tests prove the source's renormalization-group, contact-model, or asymptotic-matching inputs. The final lattice prefactor is recovered only conditionally on those inputs, in one original lattice time step. Relevance is L4/5, separately from realized contribution. Contribution assessment: C2/5, explicit local error certificates and spectral bounds are proved using classical tools, without closing the global lattice-mass dependencies.

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1 Scope, Source, and Status of Claims

The assigned source is *Exact mass asymptotics for the two-dimensional $O(4)$ lattice model*, dated October 5, 2026, by OpenAI [1]. The reference pin is

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

This report distinguishes three logically different statements:

1. *Source statements*: mathematical claims actually made in the pinned manuscript, including its long RG, Bethe, and endpoint arguments. Reading them is not certifying them.
2. *Proved local results here*: Sections 4–5 and 6–7, with the displayed hypotheses. Their proofs use positive Neumann series, finite Taylor moments, Tonelli's theorem, and elementary positive spectral moments.
3. *Conditional consequences*: the lattice interpretation and prefactor in Section 8, which retain the source's imported and internal large-scale dependencies.

The complete assigned mathematical source was read: all seven main sections, both nested Bethe sections, the comparison figure, the main file, and both bibliographies. Companion manuscripts cited by the source were not fully read or independently validated here. Private-corpus search used existing local maps and targeted actual files; there is no claim to have read the entire private corpus. Precise private passage scopes and hashes, public pinned-byte comparisons, finite checks, and build/visual records accompany this report. No private material was submitted to web search or a remote computation service.

1.1 Why This Is a Local Contribution

The core source already has a positive inverse and an exponentially weighted exterior representation. We do not claim to discover either. Our addition is an explicit usable residual budget, its coupling uncertainty, and the resulting interval in scaled mass units. The trace addition improves a finite inequality, not its asymptotic order. Neither change supplies the exact RG comparison or finite-circle integrable spectral theorem. The Ram interpretation is a finite measurement and

stability formulation, not a new physical regularization, a generic solution of ill-posedness, or evidence of historical novelty.

2 Normalization and Dependency Ledger

2.1 Lattice Mass and Full Observable Space

For an even square torus with side at least four, the source uses spins $\sigma_x \in S^3 \subset \mathbb{R}^4$, normalized probability surface measure ω_3, and

$$\mathrm{d}\mu_\beta = Z_\beta^{-1} \exp\left(\beta \sum_{\{x,y\}} \sigma_x \cdot \sigma_y\right) \prod_x \mathrm{d}\omega_3(\sigma_x), \qquad \beta > 0, \tag{2.1}$$

with each unoriented nearest-neighbor bond counted once. The infinite-volume state is the periodic local weak limit specified in the source. Existence/uniqueness and its admitted trajectories involve the cited companion `SharpO4`; those imports are not proved by this report. Site reflection is $t \mapsto -t$. With normalized positive self-adjoint one-step OS contraction T and vacuum Ω, the source's mass is

$$m_{\mathrm{lat}}(\beta) = -\log\|T|_{\Omega^\perp}\|. \tag{2.2}$$

It is not just a spin-sector gap, not a two-step mass, and not yet the rescaled mass m_{lat}/a. Rotation-invariant centered local observables are included. Source Section 2 proves finite-graph monomial covariance bounds of orders G^2 for even/even and G for odd/odd, with mixed parity zero, and then uses density to identify the full nonvacuum spectral edge with the spin edge. On a finite row its positive trace-class kernel is

$$K(s,s') = \exp\left(\frac{\beta}{2}V(s) + \beta\sum_i s_i \cdot s_i' + \frac{\beta}{2}V(s')\right).$$

For $U = K/\lambda_1$, $\tau = \lambda_2/\lambda_1$, the even centered norm is at most τ^2. The odd detector has overlap at least $\tau/(1+\tau)$; its visibility constant may depend on fixed β. Uniformity in β cannot be silently inferred from that detector argument.

2.2 Wilson Scale and Quantifier Order

In source `prop:wilson-input`, first fix a sufficiently large dyadic L, then a sufficiently large H. With $H_j = H + j\log L/\pi$, admitted terminal parameters lie in $[0.9H, 1.1H]$ and intermediate couplings in $[H_j/2, 2H_j]$. The stated coverage is of every sufficiently large β, not merely a selected subsequence. For its scale $a = L^{-N}$,

$$N\log L = \pi\beta - \tfrac{1}{2}\log\beta + O_{L,H}(1), \qquad a \asymp \sqrt{\beta}\,\mathrm{e}^{-\pi\beta}, \qquad m_{\mathrm{lat}}/a \le \log 8. \tag{2.3}$$

Here a is the source's auxiliary scale. It must not be mistaken for an arbitrary physical lattice spacing. For permitted dyadic cylinder width w, its conclusions include

$$E(w) = -a^{-1}\log\tau_{\beta,w/a}, \qquad |E(w) - m_{\mathrm{lat}}/a| \le C_H w^{-1/4}.$$

The source derives analogous bounds from

$$p(t,w) = 2\log Z(t,w) - \log Z(2t,w), \quad \Delta(w) = 4\log Z(w,w) - \log Z(2w,2w),$$

using $\Delta(w) = 2p(w,w) + p(w,2w)$ and positive transfer spectra. Its smallness bootstrap and capped-rate limit remove a cap only after the imported $m_{\mathrm{lat}}/a \le \log 8$ bound. A finite norm estimate here does not establish this trajectory coverage.

2.3 Brownian and Contact Normalizations

Source Section 3 uses

$$U(2) = (\mathbb{R}/2\pi\mathbb{Z} \times SU(2))/((v, Q) \sim (v + \pi, -Q)), \quad G = \mathrm{e}^{iv}Q,$$

the unnormalized row Hamiltonian $\mathcal{E} = \frac{1}{2}(-\partial_v^2 - \Delta_{S^3})$, row trace $\operatorname{Tr} \mathrm{e}^{-W\mathcal{E}}$, and oriented vertical weight

$$\exp\left(\lambda \int \operatorname{Tr}(G_t G_{t+1}^*)\right), \qquad \lambda = b^2/2.$$

The weight can be complex; positivity needed later is obtained from the conserved-particle Gram transfer operators, not from pointwise positivity of this weight. Their direct-sum traces are finite. The associated contact Hamiltonian on symmetric H^1 wavefunctions on a circle, with four colors, is

$$H_c = -\sum_i \partial_i^2 + 2c \sum_{i<j} \delta(X_i - X_j) P_{ij}^L, \qquad c > 0;$$

the integral equations below use $c = 2$. The source's parity proposition assumes, for fixed $D > 0$,

$$W \asymp \mathrm{e}^{\pi b} b^{-1/2} (\log b)^D, \qquad |n_p/W - b^2| \le C_* b \quad \text{for every maximizing particle number } n_p. \tag{2.4}$$

It concludes $\log \tau_0 - \log \tau_1 \sim 32\sqrt{2}\sqrt{b}\, \mathrm{e}^{-\pi b}$. The word "every" prevents choosing one favorable sector while ignoring another norm maximizer.

2.4 Nonlocal Inputs Kept Visible

The source's finite-circle argument uses the line spectral resolution, exclusion of higher clusters, paired roots $u_j \pm i$, possibly one single root at zero, positive overlap with the simple ground state, and continuation in circle length. Its counting density obeys

$$d = \rho_{B_c} - \frac{\ell}{W}(I - K)^{-1}(K_1 - K_3) + O(\mathrm{poly}(b)/W^2), \qquad \ell \in \{0, 1\}.$$

Control of endpoint errors and of all maximizing sectors gives an $O(W^{-1})$ parity error. These are substantive integrability and finite-size statements, not consequences of scalar moment fitting.

Sections 5–6 of the source match four kinetic coefficients using (b, r, c_x, c_t), retain normalized insertions through order four, control complete-support interactions, and include real lifts, winding sectors, complex weights, and integer curvature. Its normalized insertion estimates are not automatic consequences of uninserted partition-function comparison. In particular, differentiating an $O(\mathrm{e}^{-c\beta})$ partition-function error twice in a twist is not justified without derivative control. The finite-volume twist comparison in Section 7 is supplied by the source's separate normalized order-two estimates. We keep all these RG and comparison premises conditional.

3 The Exact Scalar Problem Under Examination

For $B > 0$, define

$$K_a(x) = \frac{a}{\pi(a^2 + x^2)}, \quad K = K_4, \quad S(x) = \tfrac{1}{4}\operatorname{sech}(\pi x/2), \quad r(x) = \frac{2x}{x^2 + 1} + \frac{2x}{x^2 + 9}. \tag{3.1}$$

The source equations `eq:B3-B4`, with zero extension of interior functions, are

$$\rho_B = \pi^{-1} + K * \rho_B \qquad (|x| < B), \tag{3.2}$$

$$\epsilon(x) = p(x) + (K * \epsilon_{\mathrm{in}})(x), \qquad p(x) = 4\log b - \log((x^2 + 1)(x^2 + 9)), \tag{3.3}$$

$$\epsilon(B) = 0, \qquad \epsilon_{\mathrm{in}} = \epsilon \mathbf{1}_{(-B,B)}.$$

Let $\eta = -\epsilon'$ be odd on $(-B, B)$. On $[0, B]$ introduce the positive folded kernel

$$J_B(x,y) = K(x-y) - K(x+y), \qquad (J_B v)(x) = \int_0^B J_B(x,y)v(y)\,\mathrm{d}y. \tag{3.4}$$

It is nonnegative and strictly positive when $x, y > 0$. The actual half-interval problem is

$$(I - J_B)\eta = r, \qquad 4\log b(B) = \log 9 + \int_0^B a_0(x)\eta(x)\,\mathrm{d}x, \quad a_0(x) = \tfrac{2}{\pi}\arctan(4/x), \tag{3.5}$$

where $a_0(0) = 1$ by continuity. This a_0 is not the lattice scale a.

Lemma 3.1 (Elementary scalar well-posedness)**.** *For every finite $B > 0$, set*

$$\alpha_B = \tfrac{2}{\pi}\arctan(B/4) < 1, \qquad \kappa_B = 1 - \alpha_B > 0. \tag{3.6}$$

On real $C([0, B])$ with supremum norm, $\|J_B\| \le \alpha_B$ and $(I - J_B)^{-1}$ is positive with norm at most κ_B^{-1}. The unique η in (3.5) *is zero at zero and strictly positive elsewhere. Formula* (3.5) *gives a continuous strictly increasing $b(B) > \sqrt{3}$.*

Proof. The mass of $K(x-y)$ over $y \in [-B, B]$ is maximal at $x = 0$: its derivative for $x > 0$ is $K(x+B) - K(x-B) \le 0$. Its maximum is $2\arctan(B/4)/\pi$. The folded row integral is no larger than this full row mass. Thus $\sum_{j\ge 0} J_B^j$ converges in operator norm, is positive, and has norm at most $1/(1-\alpha_B)$. Since $r(x) > 0$ for $x > 0$ and $J_B(0, y) = r(0) = 0$, the assertions for η follow.

For $B_2 > B_1$, the restriction of η_{B_2} to $[0, B_1]$ solves the B_1 equation with additional nonnegative forcing from $(B_1, B_2]$. Applying the positive inverse gives $\eta_{B_2} \ge \eta_{B_1}$ there. The extra integral of $a_0\eta_{B_2}$ over (B_1, B_2) is strictly positive, so b increases strictly. Rescaling to $[0, 1]$, kernels and forcing depend continuously on B in supremum norm, and their inverses are uniformly bounded on compact B-intervals. This proves continuity. Positivity of the integral gives $b(B) > \sqrt{3}$. This proof alone does not establish the large-B asymptotic. □

Define $\epsilon(x) = \int_{|x|}^B \eta(y)\,\mathrm{d}y$ inside the interval and extend by (3.3). The vanishing endpoint values remove differentiation boundary terms. Evaluation at zero yields precisely (3.5); hence this construction solves (3.3). For $z > B$,

$$-\epsilon'(z) = r(z) + \int_0^B [K(z-x) - K(z+x)]\eta(x)\,\mathrm{d}x > 0. \tag{3.7}$$

Thus $\epsilon_{\mathrm{out}} \le 0$ and is strictly negative outside the endpoints. The source's exact positive mass functional is

$$m_B = -(S * \epsilon_{\mathrm{out}})(0) = -\Big\{\log(b^2/4) - [(K_1 - K_3) * \epsilon_{\mathrm{in}}](0)\Big\} > 0. \tag{3.8}$$

The identity follows from $\widehat{K_a}(k) = \mathrm{e}^{-a|k|}$, $\widehat{S}(k) = (2\cosh k)^{-1}$, so $S * (\delta_0 - K_4) = K_1 - K_3$, and $S * p = \log(b^2/(x^2+4))$. For the latter logarithmic convolution, differentiation and the constant at infinity fix the additive constant. The positive exterior representation, not subtraction of the two terms in braces, is the basis of our enclosure.

4 Residual-to-Mass Certificates

Theorem 4.1 (Residual, coupling, and scaled mass)**.** *Fix any $B > 0$. For any real $v \in C([0, B])$, suppose a rigorous bound*

$$e = v - J_B v - r, \qquad \|e\|_\infty \le \varepsilon$$

has been supplied, and let $d = \varepsilon/\kappa_B$. *Define*

$$\log \widetilde{b} = \tfrac{1}{4}\left(\log 9 + \int_0^B a_0(x)v(x)\,\mathrm{d}x\right), \tag{4.1}$$

$$H(z) = \int_z^\infty S(y)\,\mathrm{d}y = \pi^{-1}\arctan(\mathrm{e}^{-\pi z/2}),$$

$$\widetilde{m}_B = 2\int_B^\infty H(z)\left[r(z) + \int_0^B J_B(z,x)v(x)\,\mathrm{d}x\right]\mathrm{d}z. \tag{4.2}$$

In (4.2) *the same formula for* $J_B(z,x)$ *is used for exterior* $z > B$. *Then*

$$\|\eta - v\|_\infty \le d, \tag{4.3}$$

$$|\log b(B) - \log \widetilde{b}| \le \tfrac{1}{4}A(B)d, \tag{4.4}$$

$$|\mathrm{e}^{\pi B/2}(m_B - \widetilde{m}_B)| \le \frac{4}{\pi^2}d, \tag{4.5}$$

$$A(B) = \frac{2}{\pi}\left[B\arctan(4/B) + 2\log(1 + B^2/16)\right].$$

More generally any continuous $w \ge 0$ *satisfying* $(I - J_B)w \ge |e|$ *pointwise gives* $|\eta - v| \le w$, *and one can replace* d *inside the corresponding positive integrals by* $w(x)$.

Proof. We have $\eta - v = -(I - J_B)^{-1}e$. Lemma 3.1 proves (4.3); positivity gives the barrier statement. Subtracting (4.1) from (3.5) proves (4.4), since direct integration gives $\int_0^B a_0 = A(B)$.

From (3.7) and $\epsilon(B) = 0$, Tonelli's theorem for the nonnegative exact integrands yields

$$m_B = 2\int_B^\infty S(y)\int_B^y [-\epsilon'(z)]\,\mathrm{d}z\,\mathrm{d}y = 2\int_B^\infty H(z)\left[r(z) + \int_0^B J_B(z,x)\eta(x)\,\mathrm{d}x\right]\mathrm{d}z. \tag{4.6}$$

For exterior $z > 0$, $J_B(z,x) \ge 0$ and $\int_0^B J_B(z,x)\,\mathrm{d}x \le 1$. Also $H(z) \le \pi^{-1}\mathrm{e}^{-\pi z/2}$ and $r(z) \le 4/3$, because $2z/(z^2 + c^2) \le 1/c$. These facts ensure absolute convergence for every bounded v and permit subtraction and Fubini. The difference in (4.6) is at most

$$2d\int_B^\infty H(z)\,\mathrm{d}z \le \frac{4d}{\pi^2}\mathrm{e}^{-\pi B/2},$$

which proves (4.5). Keeping the factor $w(x)$ before integration proves its weighted version. □

Corollary 4.2 (Finite interval implementation)**.** *In Theorem 4.1, additionally assume* $v \ge 0$, $\|v\|_\infty \le V$, *and* $R \ge B$. *Suppose a number* Q *approximates the scaled integral* (4.2) *with its outer upper endpoint replaced by* R, *with a rigorously certified absolute error at most* $q \ge 0$. *Put*

$$E = 4d/\pi^2, \qquad T_R = \frac{4}{\pi^2}(4/3 + V)\mathrm{e}^{-\pi(R-B)/2}.$$

Then the exact scaled mass $M_B = \mathrm{e}^{\pi B/2}m_B$ *lies in*

$$[\max\{0, Q - q - E\},\ Q + q + E + T_R]. \tag{4.7}$$

If v *is signed, the safe lower endpoint is instead* $\max\{0, Q - q - E - T_R\}$, *with the same upper endpoint and* $V \ge \|v\|_\infty$.

Proof. For $v \ge 0$ the omitted tail is nonnegative. Its absolute value after scaling is at most T_R by the proof of Theorem 4.1, with $4/3 + V$ in place of d and lower endpoint R. Add the quadrature and residual budgets. For signed v only the absolute tail estimate remains. Positivity of the true m_B permits intersection with $[0, \infty)$. □

4.1 Endpoint Uncertainty Is a Separate Variable

Equation (4.4) encloses $b(B)$, not the mass at a fixed requested b after replacing B by an approximate root. If $0 < B_- < B_+$ and certified coupling intervals prove

$$\widetilde{b}(B_-)\mathrm{e}^{A(B_-)d_-/4} < b_{\mathrm{target}} < \widetilde{b}(B_+)\mathrm{e}^{-A(B_+)d_+/4},$$

continuity and strict monotonicity give a unique selected endpoint in (B_-, B_+). A mass interval at that endpoint still requires uniform bounds over this endpoint interval, or a further proved endpoint-sensitivity estimate. No monotonicity of m_B is assumed here. The explicit exponential factor changes by $\mathrm{e}^{\pm\pi|\Delta B|/2}$; controlling that factor alone does not control the remaining B-dependence of M_B.

4.2 Absolute Versus Relative Accuracy

The bound is valid for every $B > 0$. Since

$$\kappa_B \sim \frac{8}{\pi B}, \qquad A(B) = O(\log B),$$

its worst-case errors are $O(B\varepsilon)$ in scaled mass and $O(B\varepsilon \log B)$ in log coupling. If the source's separate asymptotic

$$b(B) = \frac{B}{2} + \frac{\log(4\pi B) + 1}{\pi} + o(1), \qquad M_B \sim \frac{8}{\pi e\sqrt{B}} \tag{4.8}$$

is accepted, the relative mass budget is $O(B^{3/2}\varepsilon)$. Consequently $\varepsilon = o(B^{-3/2})$ is a sufficient residual condition for this bound to yield relative error $o(1)$. This is a conditional interpretation of a proved absolute certificate; it is not a new proof of (4.8). There is no asserted runtime improvement.

5 Ram Reformulation from Actual Private Passages

5.1 Finite Local Moments, Not an Infinite Formal Expansion

The actual main body of private Volume 1, revised Chapter 3 [2], states `thm:ram-local`, `eq:ram-integral-remainder`, and `eq:ram-remainder-bound`. For finite integer $N \geq 0$, an absolutely controlled kernel and a function smooth along every integration segment give

$$Tf(x) = \sum_{|\alpha|\leq N} \frac{(-1)^{|\alpha|}}{\alpha!} m_\alpha(x)\partial^\alpha f(x) + r_N(x), \quad m_\alpha(x) = \int \xi^\alpha k(x,\xi)\,\mathrm{d}\xi,$$

with $|r_N|$ bounded by the absolute $(N+1)$st moment times an appropriate derivative majorant divided by $(N+1)!$. The chapter explicitly separates exactness on polynomials from closure: its `thm:degree-closure` imposes degree conditions on the coefficient functions. Even positive normalized moment structure need not make an operator invertible; its example $I - xD$ kills x. These limitations are relevant here, not optional disclaimers.

For the present finite interval, put $y = x - \xi$ and $k_B(x,\xi) = J_B(x, x-\xi)\mathbf{1}_{[0,B]}(x-\xi)$. For $f \in C^{N+1}([0,B])$ and $x \in [0,B]$ the exact one-dimensional specialization is

$$J_B f(x) = \sum_{j=0}^{N} \frac{(-1)^j}{j!}\mu_j(x) f^{(j)}(x) + R_N(x), \quad \mu_j(x) = \int_0^B (x-y)^j J_B(x,y)\,\mathrm{d}y, \tag{5.1}$$

and

$$|R_N(x)| \leq \frac{\|f^{(N+1)}\|_\infty}{(N+1)!}\int_0^B |x-y|^{N+1} J_B(x,y)\,\mathrm{d}y. \tag{5.2}$$

Indeed Taylor's integral remainder on the segment joining x and y gives this expression before integration; compactness and continuity provide the majorant. This is a proved direct specialization, not an invocation of an unspecified Ram principle. It is exact for $f \in \mathcal{P}_N$, but $J_B\mathcal{P}_N$ need not lie in $\mathcal{P}_N$: even the zeroth row integral contains arctangents. On the whole line K_4 has divergent absolute first moment, so the same unqualified whole-line Taylor argument would fail.

5.2 Concrete Polynomial Measurements and Their Limits

Private TR-rev-1-old-tr250 [5], in `eq:moments` and the subsection "A concrete polynomial Ram construction," writes $m = Ec$ with $E_{ij} = \ell_i(\phi_j)$. It distinguishes a square inverse from a specified rectangular left inverse acting on compatible data. Its polynomial example uses a positive-a.e. weight on a nondegenerate finite interval: the Gram form $\int wp^2$ is positive for every nonzero polynomial, which proves unisolvence. It explicitly does not improve best approximation merely by changing coordinates.

Here, for any finite basis $\{\phi_j\}_{j=0}^N$ of $\mathcal{P}_N([0,B])$, one may determine a trial $v = \sum_j c_j\phi_j$ from an invertible measurement matrix. However, neither invertibility of that matrix nor satisfaction of finitely many projected equations proves a small *continuous* residual $v - J_Bv - r$. Theorem 4.1 requires a supremum bound or a pointwise barrier over the whole interval. Sampling a mesh without an interpolation or derivative error bound is insufficient.

Panelwise polynomial integrals can be evaluated exactly in logarithms and arctangents. With $u = y - z$, define primitives

$$I_0(u) = \tfrac{1}{4}\arctan(u/4), \quad I_1(u) = \tfrac{1}{2}\log(u^2+16), \quad I_k(u) = \frac{u^{k-1}}{k-1} - 16I_{k-2}(u) \quad (k \geq 2). \tag{5.3}$$

Then $I_k' = u^k/(u^2+16)$. Expanding $y^j = (u+z)^j$ and multiplying by $4/\pi$ integrates $y^jK_4(z-y)$ exactly on each panel. The second folded term uses center $-z$. This supplies exact analytic expressions, not automatically correctly rounded interval evaluations. Certified arithmetic and a bound between residual test points are still required in an implementation. Discontinuous panel approximants would require the same theorem on bounded measurable functions, or a separately stated extension; our main theorem deliberately uses continuous trials.

5.3 Stability Passage and Nonlinear Root Limitations

The actual private `stability.tex` [3] proves in `thm:pert` the finite matrix perturbation bound

$$\|\widehat{c} - c\| \leq \frac{\|e_{\mathrm{data}}\| + \eta_{\mathrm{mat}}\|c\|}{\gamma - \eta_{\mathrm{mat}}}, \qquad \sigma_{\min}(A) \geq \gamma > \eta_{\mathrm{mat}} \geq \|\Delta A\|.$$

Its `thm:ball` is an a posteriori contraction theorem for $F : U \subset X \to Y$ on Banach spaces, with a bounded isomorphism $B_0 : Y \to X$, $q = \sup\|I - B_0DF\| < 1$, and ball inclusion $\eta + qr \leq r$. The stated error is $\eta/(1-q)$. A rectangular left inverse may annihilate a residual and does not certify a zero of the full system. Its `prop:transfer` and `eq:resolvent` also require bounded maps on the specified spaces.

Our continuous linear certificate is a concrete instance of these classical resolvent ideas: $X = C([0,B])$, $F(v) = (I - J_B)v - r$, and the explicit contraction bound is $\alpha_B < 1$. A finite matrix condition number alone is not the continuous inverse bound κ_B^{-1}. Conversely, that explicit continuous bound means no unspecified regularization or global nonlinear theorem is needed for Theorem 4.1. Endpoint selection is treated by a monotone bracket rather than assuming that an unverified Newton iteration converges.

5.4 An Exact Bounded Example, Not a Low-Temperature Test

For every $0 < B \le 1$ and integer $N \ge 0$, take the exact Neumann trial

$$v_N = \sum_{j=0}^{N} J_B^j r \ge 0.$$

Using $\arctan t \le t$ and $\pi > 3$ gives $\alpha_B \le 1/6$ and $\kappa_B \ge 5/6$. Therefore

$$\|v_N - J_B v_N - r\| \le \frac{4}{3}\,6^{-(N+1)}, \qquad \|\eta - v_N\| \le \frac{8}{5}\,6^{-(N+1)}, \quad \|v_N\| \le \frac{8}{5}. \tag{5.4}$$

This follows by telescoping to residual $-J_B^{N+1} r$; the bounds hold for all B and N in the stated domain without a numerical fit. At $N = 5$, the solution error bound is exactly $1/29160$. The scaled mass error is at most $4/(9 \cdot 29160)$ using $\pi^2 > 9$. This is a rigorous small-endpoint illustration of the certificate, not a test of $B \to \infty$, not a proof of the lattice asymptotic, and not an implemented algorithm with measured cost.

6 Positive Trace Moments and Sharper Finite Bounds

Let A be a nonzero positive compact operator on a Hilbert space, and let an integer $n \ge 1$ satisfy A^n trace class. Write

$$Z = \operatorname{Tr} A^n > 0, \quad Q = \operatorname{Tr} A^{2n} > 0, \quad \tau = \|A\| > 0, \quad D = 2\log Z - \log Q, \quad c = \log Z - n \log \tau. \tag{6.1}$$

The same definitions apply to a direct sum when its total nth trace is finite. Positivity is essential: an arbitrary signed or complex trace does not define the probability weights below.

Theorem 6.1 (Two-trace refinement)**.** *Under these hypotheses,*

$$\frac{Q}{Z} \le \tau^n \le \sqrt{Q}, \qquad \frac{D}{2} \le c \le D. \tag{6.2}$$

For two such operators A_0, A_1 with the same n, define

$$R_n = \frac{\log Z_0 - \log Z_1}{n}, \qquad g = \log \tau_0 - \log \tau_1.$$

Then

$$R_n - \frac{D_0}{n} + \frac{D_1}{2n} \le g \le R_n - \frac{D_0}{2n} + \frac{D_1}{n}. \tag{6.3}$$

Proof. Let $x_j = \lambda_j(A)^n \ge 0$ and $M = \tau^n = \max_j x_j$. The spectral theorem and compactness give $Q = \sum_j x_j^2 \le M \sum_j x_j = MZ$ and $Q \ge M^2$. All sums converge. Taking logarithms proves the first assertion and $D/2 \le \log(Z/M) \le D$. Finally $ng = nR_n - c_0 + c_1$, which gives (6.3) with the indicated signs. □

The source's final trace-doubling lemma supplies $0 \le c \le D$. The added lower bound $c \ge D/2$ halves the width of the resulting *two-sector interval*: it is $(D_0 + D_1)/(2n)$ instead of $(D_0 + D_1)/n$. It does not halve the source's unknown asymptotic error constant, does not reduce the trace computation cost, and does not prove either trace's approximation to the lattice partition function.

6.1 Optimal Lower Norm Bound from Purity

The normalized eigenvalue weights $p_j = x_j/Z$ form a probability sequence. Its purity is $s = \sum_j p_j^2 = Q/Z^2 \in (0,1]$. The maximum $p = \max_j p_j$ satisfies $s \le p \le \sqrt{s}$. A stronger lower bound uses the probability constraint fully.

Theorem 6.2 (Sharp purity lower bound)**.** *For $0 < s \le 1$ set $k = \lfloor 1/s \rfloor$ and*

$$h(s) = \frac{1 + \sqrt{((k+1)s - 1)/k}}{k+1}. \tag{6.4}$$

For every probability sequence of purity s,

$$h(s) \le \max_j p_j \le \sqrt{s}, \qquad D/2 \le c \le -\log h(\mathrm{e}^{-D}) \le D.$$

The lower bound is attained by k entries equal to $h(s)$ and one remaining entry $1 - kh(s)$ (zero entries may be omitted). For fixed $0 < s < 1$, $\sqrt{s}$ is a supremum of the possible maxima, approached by finite probability sequences.

Proof. For any cap $p \in (0,1]$, write $j = \lfloor 1/p \rfloor$. Convexity of x^2 shows that moving mass from a smaller nonsaturated entry to a larger one, until either the larger equals p or the smaller vanishes, does not decrease the squared sum. The maximal squared sum under the cap and total mass one is therefore

$$F(p) = jp^2 + (1 - jp)^2, \qquad p \in [1/(j+1), 1/j].$$

This also bounds infinite sequences: truncate a sequence and pack its remaining nonnegative total mass into entries bounded by p; the convex packing operation bounds each finite partial contribution and its tail, and passage to the limit preserves the inequality. Equivalently, the packed vector majorizes every such sequence. On each stated interval, $F'(p) = 2j((j+1)p - 1) \ge 0$ and F is continuous and increasing from $1/(j+1)$ to $1/j$. Its inverse at s lies in $[1/(k+1), 1/k]$ and solving $kp^2 + (1-kp)^2 = s$ gives (6.4). Thus $s \le F(p)$ implies $p \ge h(s)$. The displayed extremizer has that maximum and purity. The elementary bound $s \le p$ then gives $h(s) \ge s$ by applying it to this extremizer.

For the upper endpoint, use one weight p_N and N equal weights $(1 - p_N)/N$, where the larger root solves $p_N^2 + (1 - p_N)^2/N = s$. For all sufficiently large N, that root exists, is the maximum weight, and tends to $\sqrt{s}$. For $s = 1$ the single weight one attains both bounds. Substituting $p = \tau^n/Z$ proves the bound on c. □

6.2 Actual Ram Moment Passage and Its Boundary

Private Volume 21, Chapter 2, `finite.tex` [4], in `v21c02:moments`, separates feasibility, determinacy, finite identification, and stability. Its positive Hankel/localizing tests are necessary conditions, not unconditional recovery theorems. Its `v21c02:zero-energy` uses exact vanishing of $\int p^2 \, \mathrm{d}\mu$ to force support in the roots; `v21c02:atomic-certificate` requires the stated rank, distinct real roots, positive weights, and reproduction of all available moments. It also gives different positive measures with identical finite moments.

Here the spectral probability measure $\nu = \sum_j p_j \delta_{p_j}$ is supported in $[0,1]$ with zeroth moment one and first moment s. Its special coupling between atom position and atom weight permits the packing argument. A general measure with first moment s need not have that structure. Neither Z, Q nor the bound $h(s)$ reconstructs the full spectrum, proves a finite-rank model, or proves uniqueness of a physical sector. The finite moment viewpoint organizes the available information and its limitations; the spectral theorem and classical convexity do the mathematical work.

7 Seam Cancellation and Error Propagation

For the source's two temporal seams at lengths $s = n, 2n$, write exact partition functions P_s^+, P_s^- and parity traces

$$Z_{0,s} = \tfrac{1}{2}(P_s^+ + P_s^-), \qquad Z_{1,s} = \tfrac{1}{2}(P_s^+ - P_s^-) > 0.$$

The exact traces are real and positive. Suppose possibly complex approximations obey $|\widehat{P}_s^{\pm} - P_s^{\pm}| \le \delta_s^{\pm}$. Then

$$\widehat{Z}_{\ell,s} = \Re\tfrac{1}{2}(\widehat{P}_s^+ + (-1)^\ell \widehat{P}_s^-), \quad \varepsilon_s = \tfrac{1}{2}(\delta_s^+ + \delta_s^-), \quad L_{\ell,s} = \widehat{Z}_{\ell,s} - \varepsilon_s, \quad U_{\ell,s} = \widehat{Z}_{\ell,s} + \varepsilon_s \tag{7.1}$$

enclose $Z_{\ell,s}$. In applications require all $L_{\ell,s} > 0$; otherwise a finite logarithmic certificate has not been obtained. Mere small relative error in P_s^+ is insufficient when the odd difference nearly cancels.

Corollary 7.1 (Certified spectral gap from seam intervals)**.** *If* (7.1) *are valid positive intervals for traces of the same two positive operators at powers $n, 2n$, then*

$$\begin{aligned} \frac{1}{n}\left[\log\frac{L_{0,2n}}{U_{0,n}} - \frac{1}{2}\log U_{1,2n}\right] &\le g \\ &\le \frac{1}{n}\left[\frac{1}{2}\log U_{0,2n} - \log\frac{L_{1,2n}}{U_{1,n}}\right]. \end{aligned} \tag{7.2}$$

Proof. Theorem 6.1 and positivity give

$$L_{\ell,2n}/U_{\ell,n} \le \tau_\ell^n \le \sqrt{U_{\ell,2n}}.$$

Subtract the logarithms in the correct sector order and divide by n. An inconsistent collection of input intervals can yield a reversed interval; that is a failed certificate, not evidence for a reversed physical mass. □

If a proved comparison supplies $|g - m_{\mathrm{lat}}| \le a\omega$, then m_{lat} lies in the interval (7.2) enlarged by $a\omega$ at each endpoint. Without such a bridge it is only a statement about the two specified operators. The difference g for arbitrary positive operators can be negative; we do not impose a physical sign on it without a physical premise.

The source's seam error is controlled relative to an untwisted positive reference. If an odd trace is as small as e^{-Ct} times that reference, an $\mathrm{e}^{-c\beta}$ reference-relative error becomes at worst $O(\mathrm{e}^{-c\beta+Ct})$ relative to the odd trace. Its hierarchy $t = o(\beta)$ makes this vanish. This is the precise reason that uncontrolled subtraction of two large seam partition functions is not acceptable. Similarly a common bulk factor must be multiplicative across lengths: if $Z'_{\ell,n} = d_n Z_{\ell,n}$ and $d_{2n} = d_n^2$, then $D'_\ell = D_\ell$ and the common factor cancels from (7.2). Arbitrary independent normalizations at the two lengths invalidate trace doubling.

8 Conditional Recovery of the Lattice Prefactor

8.1 Endpoint and Integrability Premises

The source's endpoint analysis uses the half-line factor

$$C(s) = \frac{\sqrt{s}\,\exp\{\frac{s}{2\pi}(\log(s/(2\pi)) - 1)\}}{\Gamma(1 + s/(2\pi))},$$

with $\mathcal{L}U(s) = \sqrt{\pi}/(2sC(s))$, $\mathcal{L}V(s) = \sqrt{\pi}C(s)/(2s)$ and $C(ik)C(-ik) = 1 - \mathrm{e}^{-|k|}$. The logarithmic term in the endpoint expansion is essential. Its uniform matching estimates, including endpoint residual barriers, yield (4.8) and

$$2\int_{-B}^{B} \rho_B = \frac{B^2}{4} + \frac{B}{\pi}\log B + O(B).$$

This report does not replace that Wiener–Hopf and uniform matching proof. Given its stated $o(1)$ control in $b(B)$, simple algebra gives

$$\mathrm{e}^{-\pi B/2} = 4\pi B e\,\mathrm{e}^{-\pi b}(1 + o(1)), \qquad \frac{8}{\pi e\sqrt{B}}\mathrm{e}^{-\pi B/2} \sim 32\sqrt{2}\sqrt{b}\,\mathrm{e}^{-\pi b}.$$

Using only $b = B/2 + O(\log B)$ would not identify this constant.

8.2 Coupling Shift and Helicity Signs

In the source, the lattice–Brownian helicity comparison takes the iterated limit

$$\lim_{m\to\infty} \limsup_{s\to\infty} \limsup_{\beta\to\infty} |h_i^B(mL^s) - h_i^F(mL^s)| = 0.$$

The inner limit keeps m, s fixed. It is not a statement uniform in arbitrary growing m, s. Write $D_i = 4\sin^2(k_i/2)$, $D = D_x + D_t$, and $D_* = k_x^2 + D_t$. The source obtains finite-volume expansions

$$\begin{aligned}
h_i^F &= \beta - \tfrac{3}{2}\int_{W,n} \frac{D_i}{D} - 4\int_{W,n} \frac{\sin^2 k_i}{D^2} + o(1),\\
h_x^B &= b/r - 4\int_{B,n} \frac{k_x^2}{D_*^2} + o(1),\\
h_t^B &= br - 2\int_{B,n} \frac{D_t}{D_*} - 4\int_{B,n} \frac{\sin^2 k_t - D_t^2/4}{D_*^2} + o(1).
\end{aligned}$$

These W, n and B, n subscripts denote the source's Wilson and Brownian momentum sums, not the endpoint B. The complex-current product is bilinear, not a conjugated variance: the iT term contributes a minus sign. Averaging the directions uses $r = 1 + O(\beta^{-1})$, $b = \beta + O(1)$ and hence $b(r + r^{-1})/2 = b + o(1)$. With the same infrared cutoff in both propagator terms, the source's finite integrals include

$$\int \frac{D_t}{D_*} = \frac{2}{\pi}, \quad \int \frac{D_t^2}{D_*^2} = \frac{1}{\pi}, \quad \frac{1}{2}\int \frac{D_x^2 + D_t^2}{D^2} = \frac{1}{2} - \frac{1}{2\pi},$$

and the matched propagator difference is $\log 2/(4\pi)$. The resulting source shift is

$$\beta - b = d_0 + o(1), \qquad d_0 = \frac{1}{4} - \frac{1 + \log 2}{2\pi}. \tag{8.1}$$

It is scheme-specific. A mass-to-Λ ratio in a continuum scheme cannot simply be substituted for it. We have checked the final prefactor algebra; the RG premises needed to justify these expansions remain conditional.

8.3 Scale Hierarchy and Final Implication

At the source's terminal rectangles, up to its dyadic choices,

$$w \asymp \beta\log^2\beta, \quad t \asymp \sqrt{\beta}\log^4\beta, \quad X = w/a, \quad n = t/a, \quad W = rX/b. \tag{8.2}$$

Thus $t = o(\beta)$, $t = o(w)$, $\beta w/t \gg t$, $\beta t/w \to \infty$, and $w/t = o(t)$. Together with (2.3), W has the Brownian size in (2.4), with logarithmic power two. The extra circle determinant has doubling

contribution $O(1 + w/t + \log\beta + \log(w/t)) = o(t)$. Temporal winding errors $\mathrm{e}^{-c\beta w/t}$ remain negligible after the odd-sector loss e^{Ct}; spatial twists and windings have their own estimates. The scalar comparison cannot be applied before these cancellations and error budgets are controlled.

Subject to those premises, the source proves

$$\log(Z_{1,n}/Z_{0,n}) = -nm_{\mathrm{lat}} + o(t), \qquad D_\ell = 2\log Z_{\ell,n} - \log Z_{\ell,2n} = o(t).$$

Theorem 6.1 then gives $g = m_{\mathrm{lat}} + o(a)$ since $n = t/a$. The source's nearby-fugacity comparison uses $b' = b\mathrm{e}^{\sigma/(2b)}$, $r' = rb'/b$ to keep heat length fixed while changing fugacity, and proves the density window for *all* maximizing sectors. Assuming that step, the integrable parity result, and (8.1), one obtains

$$\begin{aligned} m_{\mathrm{lat}} &\sim 32\sqrt{2}\sqrt{b}\,\mathrm{e}^{-\pi b} \\ &\sim 32\sqrt{2}\,\mathrm{e}^{\pi d_0}\sqrt{\beta}\,\mathrm{e}^{-\pi\beta} = 32\mathrm{e}^{\pi/4-1/2}\sqrt{\beta}\,\mathrm{e}^{-\pi\beta}. \end{aligned} \tag{8.3}$$

The equality uses $\sqrt{2}\exp(-(\log 2)/2) = 1$. The additive $o(a)$ is relatively negligible only because (2.3) makes a comparable to the claimed leading scale. This paragraph is a conditional implication, not an independent proof of the source's global theorem.

9 Primary Literature and Source Comparison

The 1990 publisher record for Hasenfratz, Maggiore, and Niedermayer [6] identifies the exact $O(3)$ and $O(4)$ two-dimensional continuum mass-gap calculation through Bethe ansatz and perturbative matching in the modified-minimal-subtraction scheme. The accessible publisher abstract and metadata were checked; the full article was not independently reviewed here. Thus this citation supplies historical method and normalization context, not verification of the lattice prefactor in (8.3).

Tracy and Widom's actual 2016 primary PDF [7], opening mathematical pages, writes the scaled Cauchy operator

$$(\mathcal{L}_\kappa f)(x) = \frac{\kappa}{\pi}\int_{-1}^{1} \frac{f(y)}{(x-y)^2+\kappa^2}\,\mathrm{d}y.$$

It distinguishes the singular small-κ Bose equation $f - \mathcal{L}_\kappa f = 1$ from the Fermi equation $f + \mathcal{L}_\kappa f = 1$. Its stated rigorous Fermi energy estimate cannot be transferred unchanged to the source's minus-sign equation with $\kappa = 4/B$. This comparison supports the need for endpoint analysis; a familiar kernel does not establish the source's particular forcing, edge constants, or contact-model spectral identification.

A current bounded primary-literature check also examined the opening mathematical discussion of Mariño, Miravitllas, and Reis (2025) [8]. Its claims about convergence of partial trans-series in integrable asymptotically free models are described there as numerical evidence, while reorganizations of formal trans-series retain assumptions about summation and possible further sectors. That evidence is not a proof of the present lattice RG matching. This is a targeted literature check through October 8, 2026, not an exhaustive claim of priority or absence of related results.

Mathematical issue	Source/private starting point	Realized addition and remaining limit
Scalar inverse	Source `lem:brownian-equations`: positive inverse, $O(1+B)$ bound. Ram stability: a posteriori residual control.	Explicit κ_B, coupling interval, positive scaled-tail residual and truncation budgets. No new endpoint asymptotic.
Finite moments	Ram local Taylor theorem with absolute remainder and separate closure conditions.	Actual Cauchy-folded moments and exact primitive recurrence. No closure claim or finite-matrix-to-continuous shortcut.
Trace extraction	Source's $0 \le c \le D$ in final assembly. Ram finite moments separate feasibility from identification.	$D/2 \le c \le D$, sharp purity lower norm bound, explicit seam intervals. No reconstruction of full spectrum.
Full lattice mass	Source's Wilson, Bethe, RG, twist, and density-window arguments plus companion inputs.	Normalization and conditional prefactor derivation made explicit. Those dependencies are not discharged.

10 Verification Scope and Limitations

The accompanying author check script uses exact rational arithmetic to test the positive two-trace inequalities on 984 bounded spectrum/power cases and their two-sector interval consequences on 4,408 deterministic pairs, each also checked with three uncertainty levels. It checks 3,239 rational cap-packing extremizer identities, 21 small-B Neumann constants, and 18 symbolic primitive/area/tail/prefactor identities. The primitive degrees are zero through fourteen. These tests check finite algebra; the general propositions are justified by the proofs above, not extrapolation from the tests. The elementary logarithmic prefactor identity is checked symbolically/algebraically, not by agreement of decimal constants.

No numerical solution of the private corpus or manuscript is used as empirical proof. In particular, no computed residual is advertised as a supremum certificate without a between-node bound, no floating-point integral is called a rigorous enclosure, and no finite spectrum certifies an infinite-volume phase. Public source bytes are compared to the requested pin independently of a Git checkout. Private sources are hashed only locally. Evidence records distinguish full assigned-manuscript reading from selected private passages and selected public PDF pages.

10.1 Independent Review: A Bounded Conditional Pass

The completed separate source review, `_audit/M389-independent-source-review.txt`, was read in full by this author. It reports no blocking error in the checked final mass-constant implication, not a global theorem certification. Its full reads cover the introduction, Wilson argument, final matching/assembly, main file, and bibliographies. Its other reads are selected ranges of the Brownian, Bethe, Bethe-transfer, RG, and comparison files; it did not read the mathematical content of the nested Bethe-line file. Its source-PDF visual review covers pages 3–5 and 98–108 of 109 pages. This differs from the author's complete assigned-source reading and must not be merged into a fictitious complete independent audit.

The reviewer explicitly leaves the full exact-RG initialization/localization, endpoint matching, finite-circle root construction, line spectral resolution, and companion foundations unverified. Its determinant, Gaussian-sign, and finite positive-spectrum checks are bounded algebra checks.

Its floating momentum sums have no interval enclosure and prove no asymptotic rate. We retain these boundaries. In particular, spatial winding suppression alone would not resolve the odd trace: the source also needs its stronger spatial quaternion seam estimate. Its Gaussian zero-mode product is n^2, and the factor $\sqrt{nX}$ cancels the dangerous fine-volume logarithm to leave $\sqrt{X/n}$. The review checks these mechanisms conditionally on their upstream inputs. Its source-only remit neither validates this report's new extensions nor assigns their contribution score; the parent's separate assembled-source read found no substantive local error in those proofs.

10.2 Pros and Cons

The residual certificate has an explicit domain and a positive error propagation path. It avoids subtractive cancellation in the scalar mass representation and separates discretization, residual, quadrature, tail, and endpoint uncertainty. Its bound may be pessimistic: κ_B^{-1} grows linearly, the exterior row bound by one discards decay, and a relative mass certificate needs increasing residual accuracy. A weighted barrier can improve the bound only after its inequality is proved.

The two-trace bounds use exactly the lengths already present in the source and are insensitive to a correctly matched common bulk normalization. Their limitations are equally concrete: they require positivity and trace-class powers, small odd-sector absolute errors, and the same transfer operator at both lengths. Sharper finite trace extraction does not settle the difficult matching or maximize-over-particle-number steps.

10.3 Future Work With Testable Deliverables

One next step is an interval-arithmetic implementation of the panel recurrence (5.3), with a verified residual derivative bound between nodes and a complete tail budget. A second is to construct source-adapted positive barriers that exploit the exterior Cauchy kernel rather than the norm bound by one. A third is an endpoint-sensitivity theorem yielding a mass interval uniformly over a certified B bracket. Each is future work, not contribution credited here.

On the source side, a complete independent audit would have to validate the admitted-trajectory coverage, the normalized insertion estimates, the finite-circle ground-state identification, and the uniform endpoint matching, including all invoked companion results. No amount of polynomial coordinate reformulation substitutes for those tasks. Additional trace moments might narrow norm intervals, but the cost, conditioning, and physically accessible measurements would need separate analysis before any efficiency claim.

11 Conclusion and Assessment

Relevance L4/5 reflects direct contact with the manuscript's scalar mass functional and final parity extraction. Contribution C2/5 reflects the smaller realized outcome: proved explicit residual and trace certificates, grounded in actual private mathematical passages and classical tools, with precise conditional connections to the lattice claim. The source's headline asymptotic is neither certified nor refuted by this report. The useful deliverable is a set of local inequalities whose inputs can be checked and whose missing global premises are visible.

A Private Passage Locations and Reading Scope

All following paths are under `C:/Users/mural/OneDrive/Documents/`. Bibliography keys identify actual mathematical passages, not merely titles. Full local hashes and exact read scopes are retained in the evidence JSON.

1. [2]: `RamCalBooks/volumes/volume1-rev2/chapters/ch03-local-moments-rev2.tex`. Main chapter body read, including the finite expansion, integral remainder, degree-closure theorem and noninvertibility example. Nested inputs are not represented as read.
2. [3]: `RamNonlin-IE-PDE-astra/TR1-lin-nonlin/stability.tex`. Entire file read; specific uses are `thm:pert`, `thm:ball`, `prop:transfer`, and `eq:resolvent`.
3. [4]: `RamCalBooks/volumes/volume21/chapters/ch02/finite.tex`. Entire file read, including `v21c02:moments`, `v21c02:zero-energy`, `v21c02:atomic-certificate`, and the finite null-pair example.
4. [5]: `RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`. Lines 146–190 and 441–490 read, including `eq:moments` and the concrete polynomial Gram construction. No full-manuscript reading is claimed for this private item.

References

[1] OpenAI. *Exact mass asymptotics for the two-dimensional $O(4)$ lattice model*. October 5, 2026. Pinned manuscript, Sections 1–7 including nested Bethe line/transfer sections. `https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Exact-mass-asymptotics-for-the-two-dimensional-O4-lattice-model-October-5-2026/build`.

[2] M. Subbarao, private Ram Calculus corpus, Volume 1 revised, Chapter 3 main source body. Actual finite Taylor-moment theorem, integral remainder, and closure restrictions; path and scope in Appendix A.

[3] M. Subbarao, private *RamNonlin-IE-PDE-astra*, TR1 linear/nonlinear report, `stability.tex`. Actual finite perturbation, contraction-ball and bounded-transfer results; path in Appendix A.

[4] M. Subbarao, private Ram Calculus corpus, Volume 21, Chapter 2, `finite.tex`. Positive finite-moment feasibility, identification, and stability distinctions; path in Appendix A.

[5] M. Subbarao, private `TR-rev-1-old-tr250`. Finite measurement matrix and positive polynomial Gram construction, specified passages in Appendix A.

[6] P. Hasenfratz, M. Maggiore, F. Niedermayer. The exact mass gap of the $O(3)$ and $O(4)$ non-linear σ-models in $d = 2$. *Physics Letters B* 245 (1990), 522–528. `https://doi.org/10.1016/0370-2693(90)90685-Y`. Publisher abstract/metadata checked; not a full-paper independent review.

[7] C. A. Tracy, H. Widom. On the ground state energy of the delta-function Fermi gas. *Journal of Mathematical Physics* 57 (2016), 103301. `https://doi.org/10.1063/1.4964252`. Primary PDF: `https://www.math.ucdavis.edu/~tracy/selectedPapers/2010s/CV102.pdf`; opening mathematical pages checked.

[8] M. Mariño, R. Miravitllas, T. Reis. On the Structure of Trans-Series in Quantum Field Theory. *SIGMA* 21 (2025), 065. `https://doi.org/10.3842/SIGMA.2025.065`. Primary PDF: `https://www.imath.kiev.ua/~sigma/2025/065/sigma25-065.pdf`; opening formal-series discussion checked.